

MSc Data Science Project 7PAM2002

Department of Physics, Astronomy and Mathematics

DATA SCIENCE FINAL PROJECT REPORT

Project Title:

Analysis and Prediction of Diabetes Using Clinical and
Demographic Health Indicators

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

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ABSTRACT

This project presents a comprehensive data science approach to analysing and predicting diabetes using clinical and demographic health indicators derived from the Pima Indian Diabetes Dataset, provided by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), comprising diagnostic measurements from 768 female patients of Pima Indian heritage aged 21 years and above. A structured data mining pipeline was followed, encompassing data understanding, exploratory data analysis, data preprocessing, model building, hyperparameter tuning, and evaluation. Missing values, identified through the presence of physiologically impossible zero readings in key clinical features, were handled via two different approaches: substituting the median value of a given feature where values were missing, or removing that row entirely from the dataset. Four supervised machine learning algorithms were trained and evaluated — Gradient Boosting, XGBoost, AdaBoost, and Decision Tree — with model performance assessed using accuracy, precision, recall, F1-score, and ROC-AUC metrics, placing particular emphasis on recall given the medical significance of minimising false negatives in diabetes diagnosis. Hyperparameter optimisation was conducted using GridSearchCV with stratified five-fold cross-validation, with AdaBoost emerging as the best-performing model, achieving an accuracy of 84.8% and an F1-score of 0.76. Feature importance analysis confirmed that glucose level is the strongest physiological predictor of diabetes, followed by BMI and age, demonstrating that ensemble learning methods hold genuine potential as supportive tools in clinical decision-making for early diabetes detection.

Table of Contents

DECLARATION STATEMENT	ii
ABSTRACT	iii
1. INTRODUCTION.....	1
2. LITERATURE REVIEW.....	2
2.1 Tasin et al. (2023) — Diabetes Prediction Using Machine Learning and Explainable AI Techniques	2
2.2 Iparraguirre-Villanueva et al. (2023) — Application of Machine Learning Models for Early Detection and Accurate Classification of Type 2 Diabetes.....	2
2.3 Nguyen et al. (2023) — The Utilisation of Machine Learning Algorithms for Assisting Physicians in the Diagnosis of Diabetes	2
2.4 Agliata et al. (2023) — Machine Learning as a Support for the Diagnosis of Type 2 Diabetes	3
2.5 Kavakiotis et al. (2017) — Machine Learning and Data Mining Methods in Diabetes Research	3
2.6 Summary and Justification of Algorithm Choices	3
3. DATA.....	4
3.1 Source and Background	4
3.2 How the Data Was Collected	4
3.3 Why This Dataset Was Selected.....	4
3.4 Dataset Structure and Features.....	4
3.5 Class Distribution.....	5
3.6 Exploratory Data Analysis	5
3.7 Data Preprocessing.....	8
4. ETHICAL CONSIDERATIONS	8
4.1 Nature of the Data	8
4.2 Data Privacy and Personal Data Protection	8
4.3 GDPR Compliance.....	8
4.4 Bias and Fairness	8
5. METHODOLOGY	9
5.1 Overview.....	9
5.2 Model Selection and Architecture.....	9
5.3 Model Training.....	9
5.4 Model Evaluation.....	9
5.5 Hyperparameter Tuning	9
5.6 Feature Importance and Learning Curves	9
6. RESULTS.....	10
6.1 Baseline Model Performance	10
6.2 Confusion Matrix Analysis	10
6.3 Hyperparameter Tuning Results	11
6.4 Feature Importance	12
	iv

6.6 Learning Curve Analysis.....	13
7. ANALYSIS AND DISCUSSION	14
7.1 Model Performance Interpretation	14
7.2 Clinical Significance	14
7.3 Comparison with Literature	14
7.4 Limitations	15
8. CONCLUSION	16
9. REFERENCES.....	17
10. APPENDICES	18
Appendix A: Dataset Summary Statistics	18
Appendix B: Zero Value Counts Per Feature.....	18
Appendix C: Hyperparameter Grids Used in GridSearchCV	18
Appendix D: Python Libraries Used.....	19
Appendix E: Full Project Code	19

1. INTRODUCTION

Diabetes mellitus is a chronic condition characterised by elevated blood glucose levels due to impaired insulin function. It is a major global health concern, affecting approximately 537 million adults in 2021 and projected to rise to 783 million by 2045 (International Diabetes Federation, 2021). If unmanaged, diabetes can lead to severe complications such as cardiovascular disease, kidney failure, and blindness, placing significant pressure on healthcare systems worldwide.

The increasing availability of digital health data has enabled the application of machine learning techniques to clinical problems. Unlike traditional statistical methods, machine learning models can identify complex patterns in high-dimensional data, making them well-suited for disease prediction tasks.

This project focuses on predicting diabetes using the Pima Indian Diabetes Dataset, which contains clinical measurements from 768 female patients aged 21 and above. The dataset is widely used in medical machine learning research due to the high prevalence of Type 2 diabetes in this population.

The central research question is: Which physiological factors most strongly influence the likelihood of developing diabetes?

To address this, the project aims to:

- analyse the dataset and identify patterns,
- determine key predictive features,
- preprocess the data,
- develop and evaluate machine learning models,
- and interpret the results for clinical relevance.

A structured data mining pipeline was followed, including exploratory data analysis, preprocessing, model training, and evaluation. Four classification models — Gradient Boosting, XGBoost, AdaBoost, and Decision Tree — were applied and compared using multiple performance metrics.

The remainder of this report is organised as follows. Section 2 reviews related literature, Section 3 describes the dataset, Section 4 outlines ethical considerations, Section 5 presents the methodology, Section 6 reports results, Section 7 discusses findings, and Section 8 concludes the study.

2. LITERATURE REVIEW

The use of machine learning for diabetes prediction has attracted growing academic interest over the past decade, with researchers exploring a wide variety of algorithms, datasets, and evaluation strategies. This section reviews five peer-reviewed studies relevant to the present project, examining their methodologies and findings and drawing direct comparisons with the approach taken here.

2.1 Tasin et al. (2023) — Diabetes Prediction Using Machine Learning and Explainable AI Techniques

Tasin, Nabil, Islam and Khan (2023) explored diabetes prediction using a combination of machine learning algorithms and explainable artificial intelligence techniques, with their work published in Healthcare Technology Letters. The study was conducted on the Pima Indian Diabetes Dataset, the same dataset used in this project, and applied classifiers including Random Forest, Support Vector Machine, and Logistic Regression. What distinguished this study from many others in the field was its integration of SHAP (SHapley Additive exPlanations) values to interpret model outputs and rank the contribution of individual features to predictions. The analysis confirmed that glucose level, BMI, and age were the most influential predictors of diabetes, a finding directly consistent with the feature importance results obtained in the present project using AdaBoost.

The present study shares a common dataset and a similar interest in identifying key risk factors, but differs in its algorithmic focus. Rather than incorporating explainability tools, this project concentrates on ensemble boosting methods and systematic hyperparameter optimisation to maximise predictive performance. The two studies can therefore be viewed as complementary, with Tasin et al. prioritising transparency and interpretability while the present work prioritises model optimisation and comparative evaluation.

2.2 Iparraguirre-Villanueva et al. (2023) — Application of Machine Learning Models for Early Detection and Accurate Classification of Type 2 Diabetes

Iparraguirre-Villanueva, Espinola-Linares, Flores Castañeda and Cabanillas-Carbonell (2023) published a study in Diagnostics examining the application of machine learning for early detection and classification of Type 2 diabetes. Their work also used the Pima Indian Diabetes Dataset and tested multiple classification models to determine which offered the best predictive accuracy. A key methodological distinction in their work was the application of the Synthetic Minority Oversampling Technique (SMOTE) to address the class imbalance present in the dataset, which the present project did not employ. Instead, this project handled class imbalance through stratified train-test splitting. Their findings favoured Random Forest as the strongest classifier, whereas the present project found AdaBoost to be the superior performer. This difference in outcomes is likely attributable to the different preprocessing strategies adopted, particularly the use of SMOTE versus row deletion for handling missing data, which can substantially influence the final dataset composition and therefore model behaviour.

2.3 Nguyen et al. (2023) — The Utilisation of Machine Learning Algorithms for Assisting Physicians in the Diagnosis of Diabetes

Nguyen, Tung, Nguyen, Le, Tran, Binh and Pham (2023) conducted a study published in Diagnostics that examined the use of machine learning to support medical professionals in diagnosing diabetes, using a clinical dataset collected from patients in Ninh Binh, Vietnam. Their study tested a broad selection of algorithms including Decision Tree, Logistic Regression, Support Vector Classifier, AdaBoost, Gradient Boosting, Random Forest, and K-Nearest Neighbours, making it the most methodologically similar study to the present project in terms of algorithm coverage. Their results indicated that Random Forest achieved the highest cross-validation score. A notable difference, however, is that Nguyen et al. evaluated their models primarily using default hyperparameter settings, whereas the present project applied GridSearchCV-based tuning to each model, making the evaluation

framework here more rigorous and the reported results more representative of each model's true potential. Both studies nonetheless arrive at a shared conclusion that ensemble methods consistently outperform individual classifiers in diabetes classification tasks.

2.4 Agliata et al. (2023) — Machine Learning as a Support for the Diagnosis of Type 2 Diabetes

Agliata, Giordano, Bardozzo and colleagues (2023) investigated the application of machine learning to Type 2 diabetes diagnosis, with their findings published in the *International Journal of Molecular Sciences*. Their study employed an artificial neural network as the primary classifier, representing a deep learning approach that differs fundamentally from the tree-based ensemble methods used in the present project. The neural network demonstrated strong performance, particularly in capturing nonlinear interactions between clinical variables, and the authors highlighted its potential value as a long-term predictive tool in diabetes management. The present project deliberately chose not to pursue neural network architectures, primarily because the relatively small dataset available after preprocessing is not well-suited to deep learning methods, which typically require substantially larger volumes of training data to generalise effectively. Ensemble methods such as AdaBoost and Gradient Boosting offer a more practical and interpretable solution for datasets of this scale.

2.5 Kavakiotis et al. (2017) — Machine Learning and Data Mining Methods in Diabetes Research

Kavakiotis et al. (2017) conducted a large-scale review of machine learning methods in diabetes research, analysing over 85 studies. They found that algorithms like Support Vector Machines and Logistic Regression were commonly used, with Logistic Regression serving as an interpretable baseline. However, ensemble methods consistently showed better generalisation performance. This supports the choice of ensemble models in the present project for improved predictive accuracy.

2.6 Summary and Justification of Algorithm Choices

The literature consistently supports ensemble learning methods for diabetes prediction, as they outperform single classifiers. Based on this, this project uses Gradient Boosting, XGBoost, and AdaBoost, with a Decision Tree as a baseline. GridSearchCV with stratified cross-validation was applied to ensure optimal performance. SMOTE was not used, as the study prioritised data integrity over sample augmentation, which is acknowledged as a limitation.

3. DATA

3.1 Source and Background

The dataset used in this project is the Pima Indian Diabetes Dataset, provided by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). It is publicly available via the UCI Machine Learning Repository and Kaggle and is licensed under CC0 (public domain), allowing free academic use: <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>

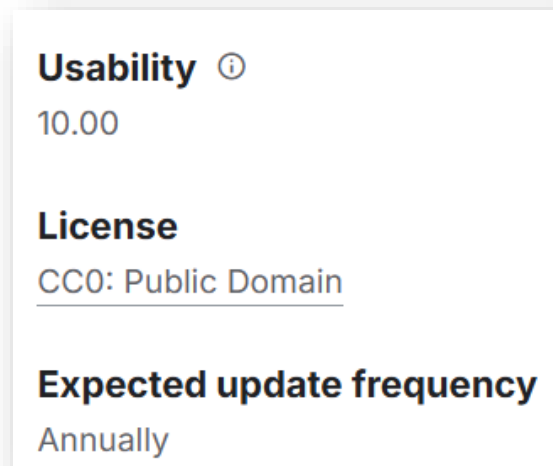


Figure 1: Kaggle dataset information including license and usability details

The dataset is licensed under CC0: Public Domain, meaning it is freely available for use without restrictions.

3.2 How the Data Was Collected

The dataset was collected from female patients of Pima Indian heritage aged 21 and above in Arizona, USA. It is widely used in diabetes research due to the high prevalence of Type 2 diabetes in this population.

3.3 Why This Dataset Was Selected

This dataset was selected due to its relevance to the research question, availability of clinically meaningful features, and suitability for machine learning analysis. It is fully anonymised and does not require ethical approval.

3.4 Dataset Structure and Features

The dataset contains 768 records and 9 columns, including 8 predictor variables and one binary target variable (Outcome).

The features include clinical indicators such as glucose, BMI, blood pressure, insulin, age, and a hereditary risk factor. The Outcome variable indicates whether a patient is diabetic (1) or non-diabetic (0).

The dataframe summary produced by `df.info()` confirmed that all 9 columns contain 768 non-null entries with no explicitly missing values recorded, with integer and float data types as appropriate. The descriptive statistics produced by `df.describe()` are presented in full in Appendix A.

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Figure 2: Tabular representation of the Pima Indian Diabetes Dataset displaying predictor variables and binary outcome label

3.5 Class Distribution

The dataset contains a moderate class imbalance, with 500 patients (65.1%) classified as non-diabetic and 268 patients (34.9%) classified as diabetic. This imbalance was acknowledged throughout the modelling process, and stratified splitting was applied when dividing the data into training and testing subsets to maintain proportional class representation in both sets.

3.6 Exploratory Data Analysis

Exploratory data analysis was carried out to understand the structure of the data, examine distributions, and identify relationships between features and the target variable. Several visualisations were produced and are presented below.

A **pie chart** was generated to examine the distribution of the target variable across the two outcome classes.

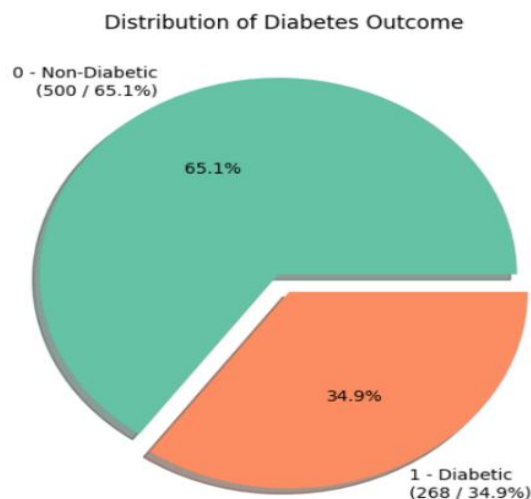


Figure 3: Distribution of Diabetes Outcome

A pie chart was used to visualise the distribution of the target variable. The dataset contains 65.1% non-diabetic cases and 34.9% diabetic cases, indicating a moderate class imbalance. This imbalance was addressed using stratified sampling during the train-test split. It is important to consider this distribution when evaluating model performance, as accuracy alone can be misleading.

Boxplots were produced to compare the distribution of key clinical features between diabetic and non-diabetic patients.

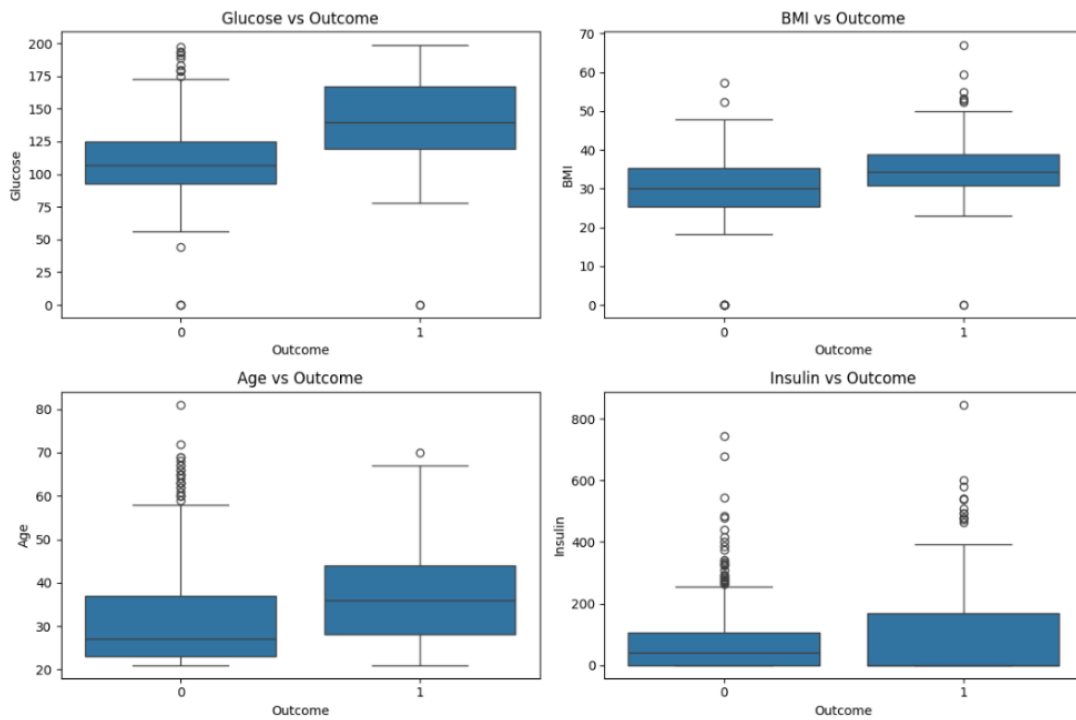


Figure 4: Boxplots — Glucose, BMI, Age and Insulin vs Diabetes Outcome

The boxplots compare the distribution of Glucose, BMI, Age, and Insulin between diabetic and non-diabetic patients. Glucose shows the strongest separation, with higher median values in diabetic patients, making it the most important predictor. BMI also shows a clear increase in diabetic cases, while Age shows a moderate difference. Insulin displays high variability and overlap, likely due to missing values.

Overall, Glucose and BMI are the most influential features. However, some overlap exists between classes, indicating that no single feature can perfectly distinguish diabetes, supporting the need for machine learning models.

A **correlation heatmap** was generated to examine linear relationships between all features in the dataset.

The heatmap above presents the Pearson correlation coefficients between all features in the dataset, including the target variable Outcome. Each cell displays the strength and direction of the linear relationship between two variables, with darker shades indicating stronger correlations. Glucose exhibits the highest positive correlation with Outcome at approximately 0.49, confirming it as the most linearly associated feature with diabetes diagnosis. BMI and Age also show moderate positive correlations with Outcome. The DiabetesPedigreeFunction shows a weak but positive correlation with Outcome, capturing the hereditary component of diabetes risk. Importantly, the correlations among predictor features themselves are relatively low, indicating minimal multicollinearity in the dataset. This is a favourable property for machine learning models, as high multicollinearity between input features can destabilise training and reduce interpretability. The heatmap therefore supports the inclusion of all eight features in model building without the need for feature elimination on multicollinearity grounds.

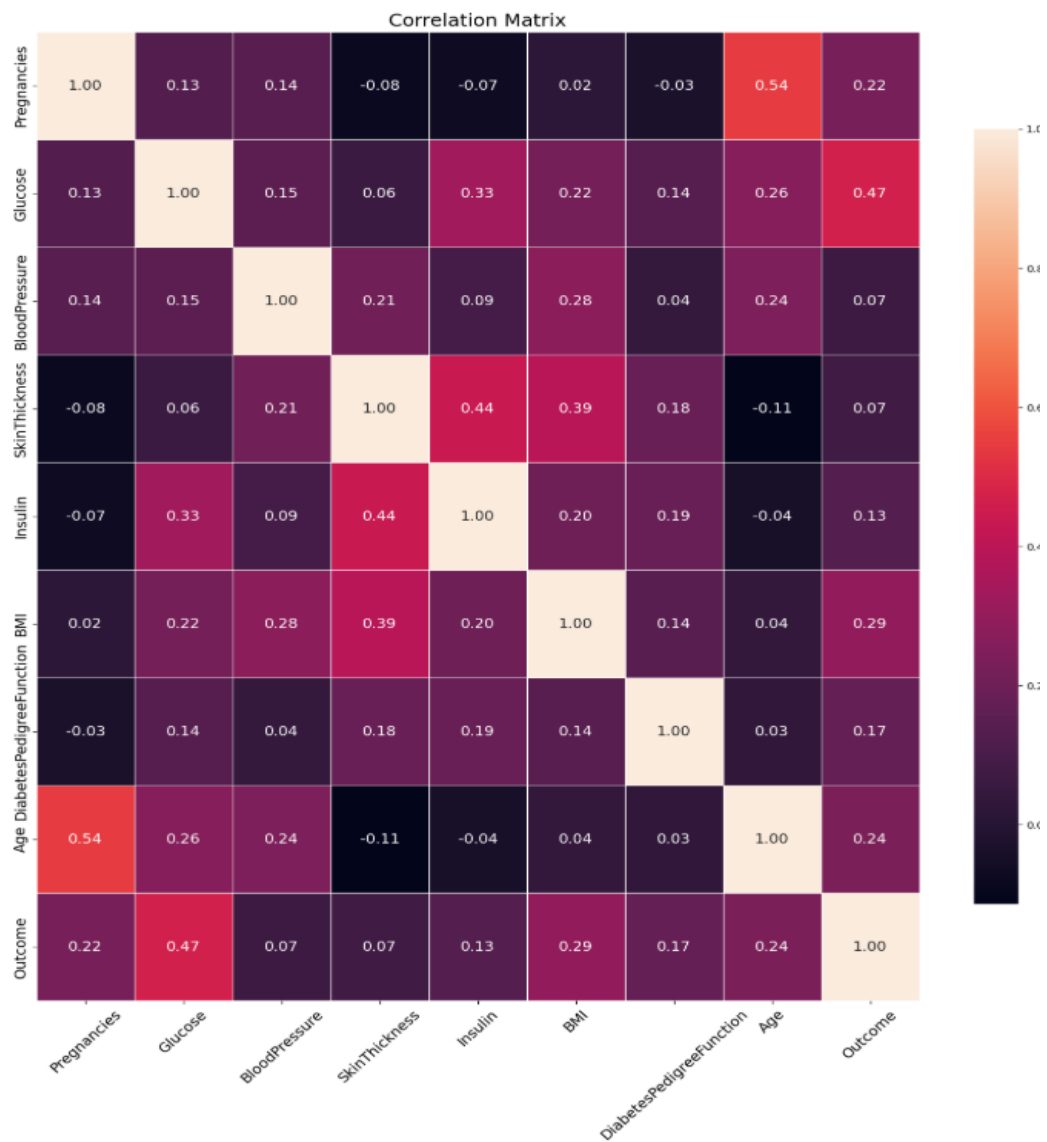


Figure 5: Correlation Matrix Heatmap

A **bar chart** was produced to visualise the count of medically invalid zero values present in each clinical feature before preprocessing.

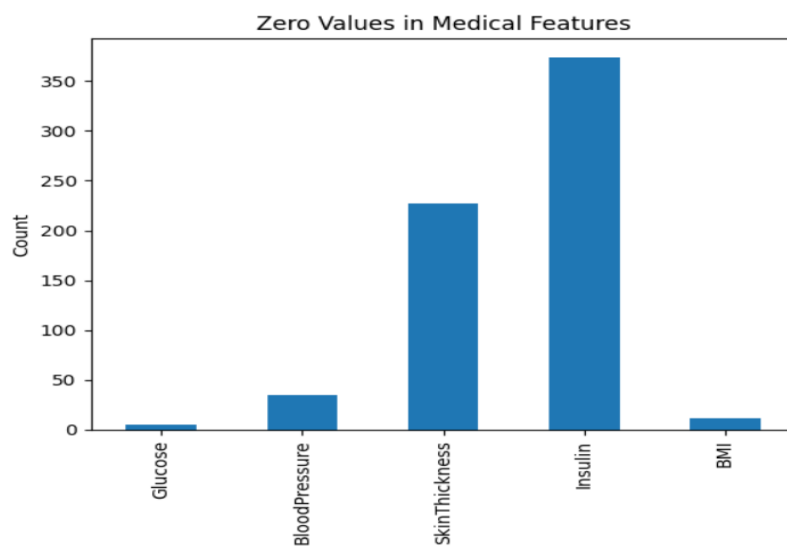


Figure 6: Zero Value Counts per Medical Feature

The bar chart shows the number of zero values in key clinical features, with Insulin and SkinThickness having the highest counts. Since zero values are medically invalid for these features, they were treated as missing data. This finding guided the data cleaning process to ensure only valid records were used for modelling.

3.7 Data Preprocessing

Data preprocessing was carried out in two stages. First, zero values in the five clinically sensitive features — Glucose, BloodPressure, SkinThickness, Insulin, and BMI — were replaced with NaN, acknowledging that these zeros represent absent rather than genuine data. Second, all rows containing NaN entries were removed using the `dropna()` function, yielding a clean dataset of approximately 392 records. While this reduces the available sample size, it ensures that every retained record contains authentic and clinically valid measurements, which is critical for producing reliable model outputs.

Following cleaning, the dataset was split into predictor features (X) and the target variable (y). A stratified train-test split of 80% training and 20% testing was applied to preserve class proportions across both subsets. Feature scaling was then performed using `StandardScaler`, which transforms each feature to have a mean of zero and a standard deviation of one. The scaler was fitted exclusively on the training data and applied to the test data separately, preventing any leakage of test set information into the training process.

4. ETHICAL CONSIDERATIONS

4.1 Nature of the Data

The Pima Indian Diabetes Dataset is a secondary, publicly available dataset released by the NIDDK. All records are fully anonymised, with no personally identifiable information. As no human participants were involved, ethical approval was not required.

4.2 Data Privacy and Personal Data Protection

Although anonymised, the dataset contains sensitive health information. It was accessed from reputable public sources and used solely for academic purposes, without sharing or commercial use.

4.3 GDPR Compliance

GDPR principles were followed throughout the project. The data was used lawfully for academic research, limited to relevant variables, and stored securely to ensure confidentiality.

4.4 Bias and Fairness

The dataset represents a specific demographic (female Pima Indian patients aged 21+), limiting generalisability. Therefore, the models are not intended for real-world clinical use without further validation on diverse populations.

5. METHODOLOGY

5.1 Overview

This section outlines the machine learning models, training procedures, and evaluation strategies used in this project. The approach follows a structured pipeline aligned with the CRISP-DM framework, focusing on model building, optimisation, and evaluation. Full implementation details are provided in the GitHub repository and Appendix E.

5.2 Model Selection and Architecture

Four supervised classification models were selected based on literature support for ensemble methods:

- **Gradient Boosting** – sequential ensemble model that minimises prediction errors using gradient descent
- **XGBoost** – optimised gradient boosting with regularisation and improved computational efficiency
- **AdaBoost** – boosting model that emphasises misclassified instances using weighted learning
- **Decision Tree** – baseline model for comparison due to its interpretability

5.3 Model Training

All four models were trained on the preprocessed training set comprising 80% of the cleaned 392-record dataset. Models were trained using unscaled features as tree-based ensemble methods are not sensitive to feature scale. The training process was implemented using the Scikit-learn library in Python, with XGBoost implemented using the XGBoost library. All models were initialised with `random_state=42` to ensure reproducibility of results.

5.4 Model Evaluation

Performance was assessed using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. Recall was prioritised due to the clinical importance of minimising false negatives.

5.5 Hyperparameter Tuning

GridSearchCV with stratified 5-fold cross-validation was used to optimise model parameters based on F1-score. The best configurations were used for final evaluation.

5.6 Feature Importance and Learning Curves

Feature importance from the AdaBoost model was used to identify key predictors. Learning curves were analysed to evaluate model generalisation and detect overfitting or underfitting.

6. RESULTS

6.1 Baseline Model Performance

Prior to hyperparameter tuning, all four models were evaluated on the test set. The results are summarised in Table 1 below.

Table 1: Baseline Model Performance Comparison

Model	Accuracy	Recall	F1 Score
AdaBoost	84.8%	73.1%	0.760
XGBoost	82.3%	76.9%	0.741
Gradient Boosting	78.5%	61.5%	0.653
Decision Tree	72.2%	65.4%	0.607

AdaBoost achieved the highest accuracy of 84.8% and an F1-score of 0.76. XGBoost followed closely with the highest recall of 76.9%, reflecting its strength in identifying true diabetic cases. Gradient Boosting achieved a reasonable accuracy but a comparatively low recall, indicating a higher rate of missed positive diagnoses. Decision Tree produced the weakest overall performance, consistent with the expectation that single non-ensemble classifiers are less effective than boosting methods on medical datasets of this nature.

6.2 Confusion Matrix Analysis

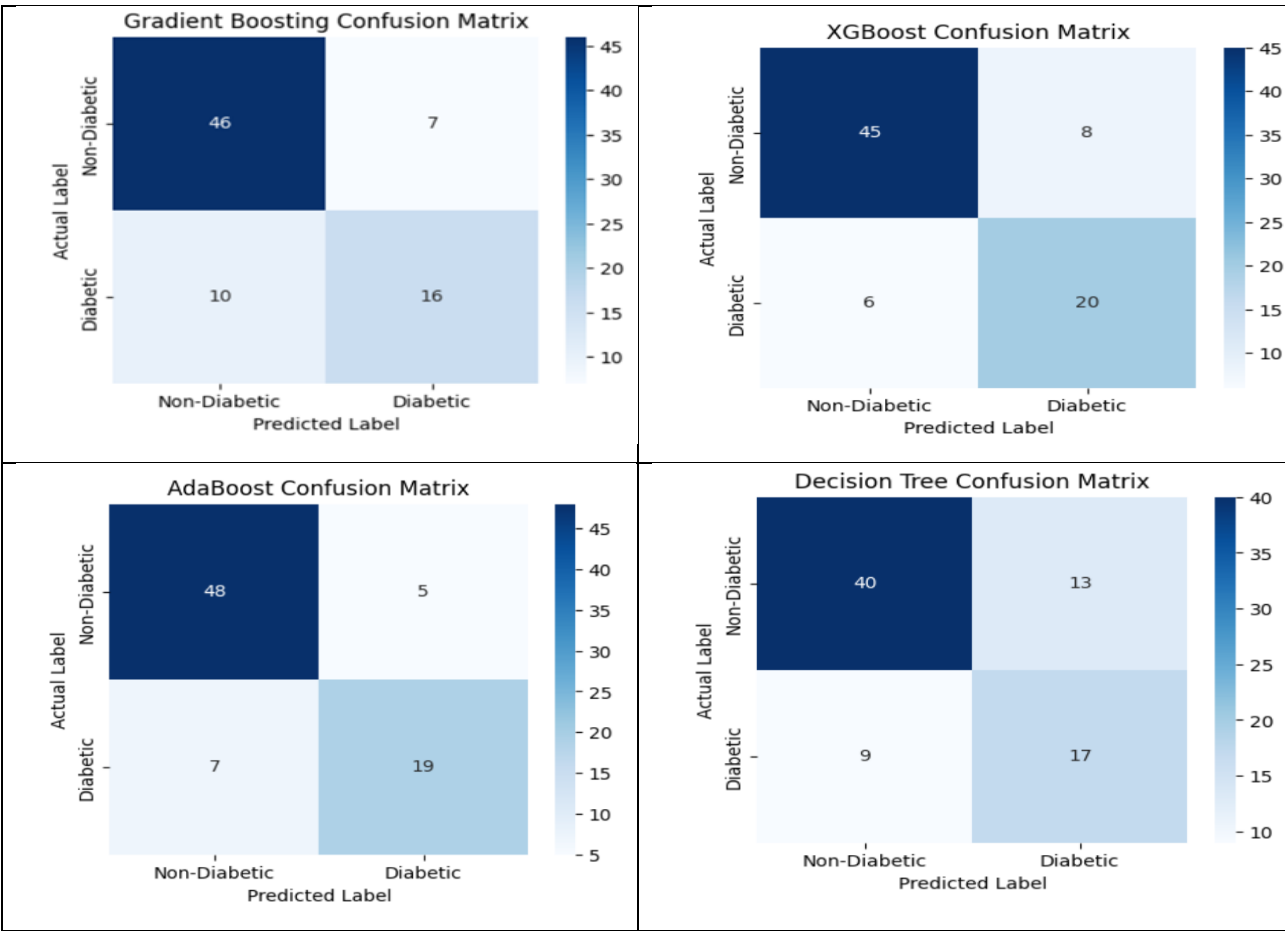


Figure 7: Confusion Matrices — Gradient Boosting, XGBoost, AdaBoost and Decision Tree

The confusion matrices above present the classification results of all four models on the test set, displaying the counts of true positives, true negatives, false positives, and false negatives for each model. In the context of diabetes prediction, false negatives carry the greatest clinical risk, as they represent diabetic patients incorrectly classified as non-diabetic, potentially leading to delayed diagnosis and serious complications. Gradient Boosting produced 10 false negatives and 7 false positives. XGBoost recorded the lowest false negative count of 6 across all models, making it the most sensitive classifier in terms of

recall. AdaBoost produced 7 false negatives and only 5 false positives, demonstrating a strong balance between precision and recall. Decision Tree generated the highest false positive count of 13, reflecting its tendency to overfit the training data and misclassify non-diabetic patients when evaluated on unseen records. Collectively, the confusion matrices confirm the superiority of ensemble boosting methods over the single Decision Tree baseline across all classification outcomes.

6.3 Hyperparameter Tuning Results

Following GridSearchCV optimisation, the best parameter sets and post-tuning performance for each model are summarised in Table 2 below.

Gradient Boosting performed best with a learning rate of 0.1, maximum depth of 3, 100 estimators, and a subsample ratio of 0.8, achieving an improved accuracy of 82.3% and F1-score of 0.731. XGBoost was optimised with a learning rate of 0.1, maximum depth of 4, 200 estimators, subsample ratio of 1.0, and column sample ratio of 0.8, but recorded a slightly lower post-tuning F1-score of 0.667, suggesting its default settings were already well-suited to this dataset. AdaBoost achieved its best results with a learning rate of 0.5 and 100 estimators, maintaining the highest post-tuning F1-score of 0.735 and accuracy of 83.5%. Decision Tree improved substantially with tuning, achieving an accuracy of 81.0% and F1-score of 0.717 using entropy criterion, maximum depth of 3, minimum samples split of 2, and minimum samples leaf of 1.

Table 2: Post-Tuning Model Performance Comparison

Model	Accuracy	Recall	F1 Score
AdaBoost	83.5%	69.2%	0.735
Gradient Boosting	82.3%	73.1%	0.731
Decision Tree	81.0%	73.1%	0.717
XGBoost	78.5%	65.4%	0.667

6.4 Feature Importance

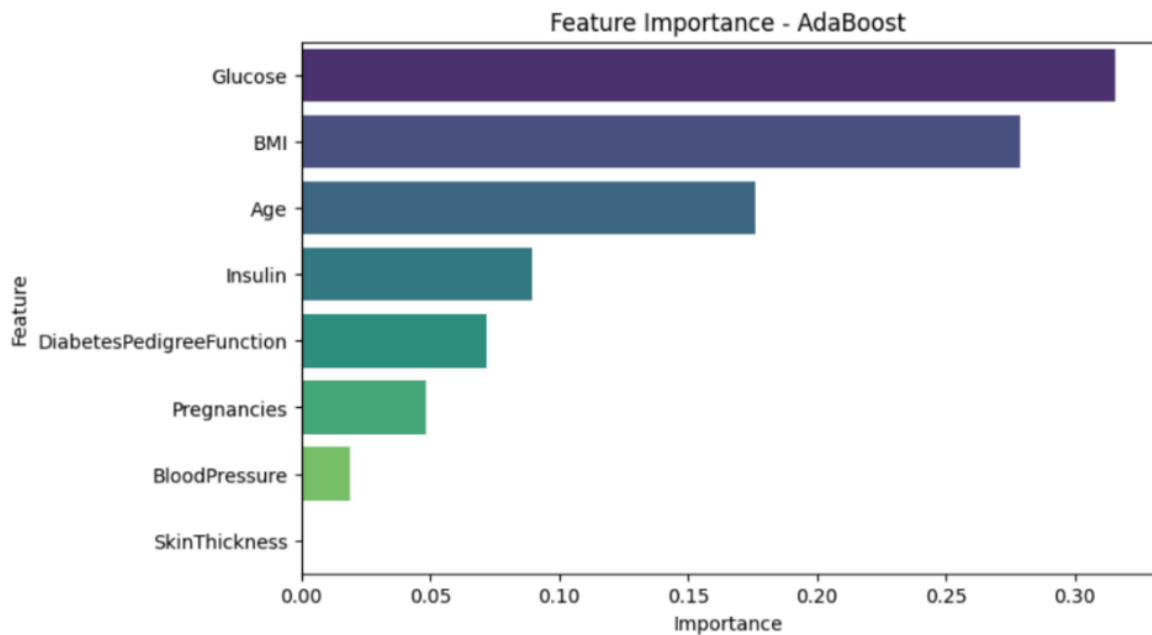


Figure 8: Feature Importance — AdaBoost Model

The feature importance bar chart above ranks all eight predictor variables according to their relative contribution to the AdaBoost model's predictions. Glucose emerges as the most important feature by a clear margin, confirming it as the dominant physiological predictor of diabetes in this dataset. This finding aligns strongly with established medical knowledge, as elevated plasma glucose concentration is the principal biomarker used in the clinical diagnosis of Type 2 diabetes. BMI ranks as the second most important feature, reflecting the well-documented relationship between obesity and insulin resistance. Age and DiabetesPedigreeFunction follow, highlighting the contributions of biological ageing and genetic predisposition to diabetes risk respectively. Insulin, BloodPressure, SkinThickness, and Pregnancies contribute relatively smaller importance scores, which may partly reflect the reduced statistical representation of these features in the cleaned dataset following row removal. This chart directly and conclusively answers the central research question of this project: glucose level is the strongest physiological factor influencing the likelihood of developing diabetes, followed by BMI and age.

6.5 ROC-AUC Analysis

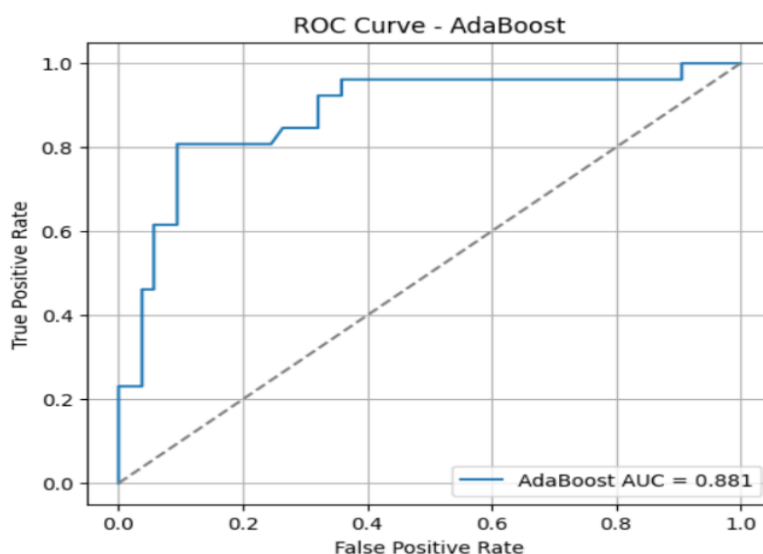


Figure 9: ROC Curve — AdaBoost Model

The ROC curve above evaluates the discriminatory performance of the AdaBoost model across all possible classification thresholds. The curve plots the True Positive Rate against the False Positive Rate, with the diagonal dashed line representing the baseline performance of a random classifier. The AdaBoost model achieved an AUC score of 0.881, indicating strong discriminatory capability well above chance level. This means the model has an 88.1% probability of correctly distinguishing a diabetic patient from a non-diabetic patient when presented with a randomly selected pair. The curve rises steeply towards the upper left corner of the plot, confirming that the model correctly identifies the majority of true diabetic cases while maintaining a low false positive rate. This metric is particularly valuable in medical classification, where the decision threshold may need to be adjusted depending on whether clinical priorities favour maximising sensitivity or specificity. An AUC of 0.881 places the AdaBoost model firmly in the category of a strong classifier and further supports its selection as the best-performing model in this project.

6.6 Learning Curve Analysis

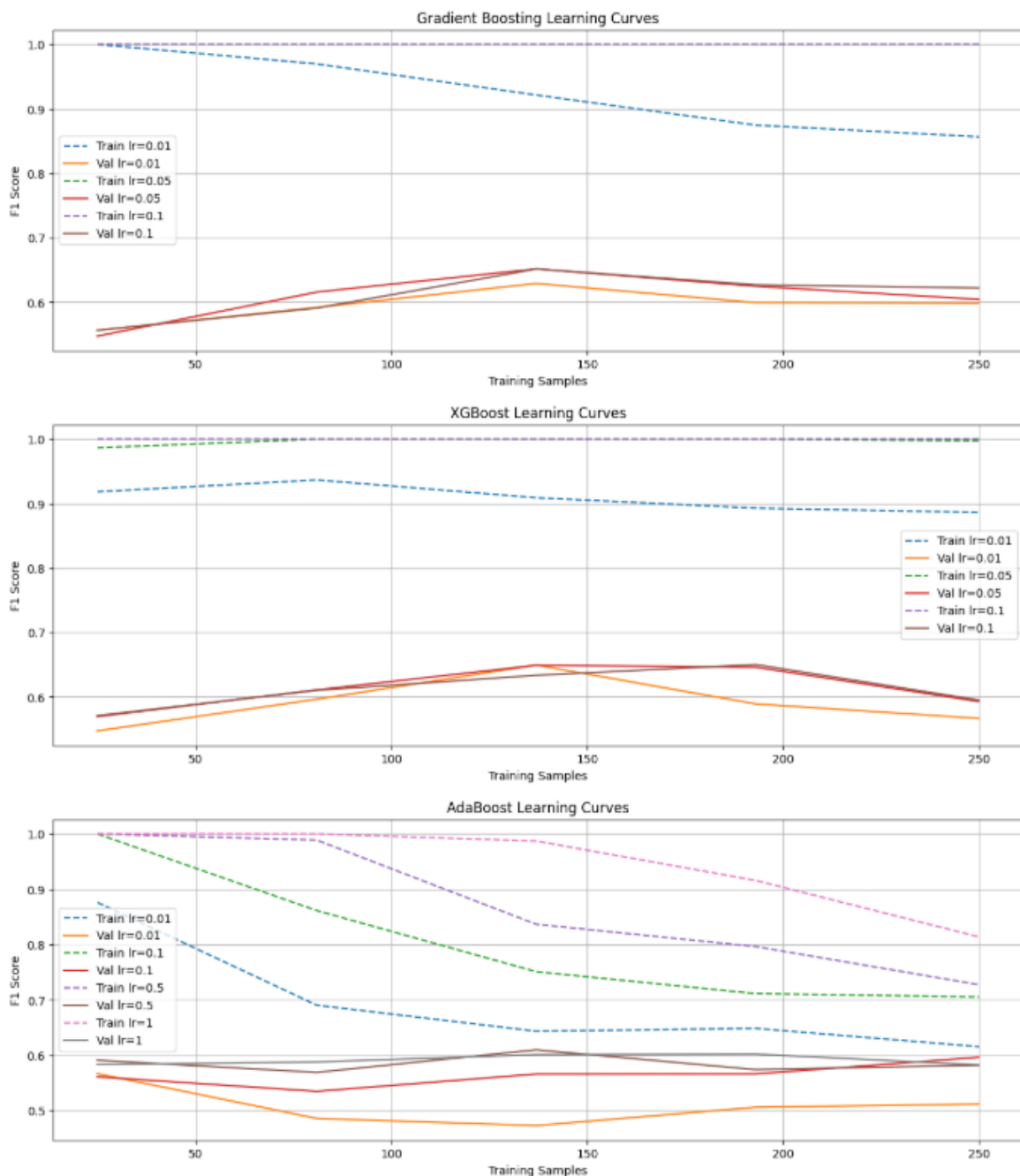


Figure 10: Learning Curves — Gradient Boosting, XGBoost and AdaBoost

The learning curves above illustrate how training and validation F1-scores evolve as the size of the training dataset increases, plotted across multiple learning rate values for each of the three boosting models. At lower learning rates such as 0.01, all three models display a wider gap between training and validation scores at smaller training sizes, which is characteristic of underfitting and high variance early in training. As the learning rate increases to 0.1, training and validation scores converge more closely as more data is introduced, indicating improved generalisation. AdaBoost demonstrates comparatively stable and consistent validation scores across different learning rates, reinforcing its robustness for this dataset. Across all three models, validation scores continue to rise without fully plateauing as training size increases, suggesting that all models could benefit from a larger training dataset.

7. ANALYSIS AND DISCUSSION

7.1 Model Performance Interpretation

The results clearly demonstrate that ensemble boosting methods outperform the single Decision Tree classifier across all evaluation metrics. This is consistent with the broader literature on machine learning for medical classification, where boosting algorithms have repeatedly been shown to deliver superior generalisation by iteratively refining predictions on difficult cases (Tasin et al., 2023; Nguyen et al., 2023). AdaBoost performed best overall in both baseline and post-tuning evaluations, which reflects its well-established effectiveness on moderately sized tabular datasets with class imbalance.

An interesting pattern emerged during hyperparameter tuning. While Gradient Boosting and Decision Tree improved noticeably, XGBoost performed slightly better in its default configuration than after tuning. This outcome is not entirely surprising given the relatively small dataset size of approximately 392 records after preprocessing. Complex hyperparameter search spaces are most effective with larger volumes of data, and on smaller datasets, aggressive tuning can result in configurations that overfit the cross-validation folds rather than generalising to the test set. This observation is worth noting for future work involving similar datasets.

7.2 Clinical Significance

From a clinical standpoint, the feature importance analysis is arguably the most meaningful output of this project. The emergence of Glucose as the dominant predictor aligns precisely with medical consensus, given that elevated blood glucose following an oral glucose tolerance test is the definitive clinical criterion for diagnosing Type 2 diabetes. The model's ability to independently identify this relationship from the data, without any embedded medical knowledge, speaks to both the quality of the dataset and the validity of the modelling approach.

BMI ranking second is equally meaningful from a preventive health perspective. It reinforces the case for lifestyle-based interventions targeting weight management as a strategy for reducing diabetes risk at a population level. The contributions of Age and DiabetesPedigreeFunction further highlight the interplay between non-modifiable risk factors such as genetics and ageing, and measurable clinical indicators such as glucose and BMI, in determining diabetes likelihood.

The relatively low importance assigned to Insulin and SkinThickness is notable and likely reflects data quality issues rather than a genuine lack of clinical relevance. Both features suffered from very high proportions of missing values, with Insulin having zero values in nearly half of all original records. Their reduced representation in the cleaned dataset may have limited the model's opportunity to learn their true predictive value.

7.3 Comparison with Literature

The results obtained in this project are broadly consistent with those reported in the reviewed literature. Tasin et al. (2023) identified the same top predictors using SHAP-based analysis on the

same dataset, validating the feature importance findings of the present study. Iparraguirre-Villanueva et al. (2023) achieved stronger recall through SMOTE-based oversampling, highlighting a genuine limitation of the present approach in addressing class imbalance. Nguyen et al. (2023) demonstrated similarly strong ensemble performance, though without systematic hyperparameter tuning, making direct comparison of absolute metrics difficult. Agliata et al. (2023) illustrated the potential of deep learning architectures, though at a computational and interpretability cost that is difficult to justify for a dataset of the size used here.

Overall, the methodology and results of this project sit comfortably within the range of performance reported by comparable published studies, and the systematic approach to tuning and evaluation represents a meaningful methodological contribution relative to several works in the existing literature.

7.4 Limitations

Several limitations of this project deserve honest acknowledgement. The demographic scope of the dataset is narrow, covering only female patients of Pima Indian heritage, which substantially restricts the generalisability of the models developed. The decision to remove rows containing invalid zero values, while preserving data integrity, reduced the working dataset by nearly half from 768 to approximately 392 records, and this smaller sample may have constrained model performance. The absence of oversampling techniques means the class imbalance was not fully corrected, which may have depressed recall scores. Finally, all models were evaluated on a single train-test split derived from one dataset, and external validation on an independent clinical dataset would be necessary before any practical application could responsibly be considered.

8. CONCLUSION

This project set out to analyse and predict diabetes using clinical and demographic health indicators from the Pima Indian Diabetes Dataset, with a specific focus on identifying which physiological factors most strongly influence the likelihood of developing the condition. Through a structured data mining pipeline encompassing exploratory analysis, preprocessing, ensemble model building, hyperparameter optimisation, and multi-metric evaluation, the project has produced results that are both technically robust and clinically meaningful.

Reviewing each objective in turn, the first objective of understanding the structure and characteristics of the dataset was fulfilled through initial data inspection, descriptive statistics, and identification of medically invalid zero values. The second objective of performing exploratory data analysis was addressed through a series of visualisations including pie charts, boxplots, correlation heatmaps, and zero value bar charts, which revealed key patterns and data quality issues. The third objective of determining the most influential physiological factors was conclusively answered through feature importance analysis, which identified glucose as the dominant predictor followed by BMI and age. The fourth objective of preprocessing the data was achieved through systematic replacement of invalid zeros with NaN values and subsequent row removal, producing a clean dataset of 392 records. The fifth objective of building and evaluating machine learning models was fulfilled through the training and comparison of four supervised classifiers across multiple performance metrics. The sixth objective of interpreting results and providing actionable medical insights was addressed throughout the results, analysis, and discussion sections.

AdaBoost consistently emerged as the best-performing classifier, achieving a baseline accuracy of 84.8% and an F1-score of 0.76, with an AUC score of 0.881 confirming strong discriminatory capability. Gradient Boosting benefited the most from hyperparameter optimisation, while Decision Tree showed substantial improvement when its depth was appropriately constrained. XGBoost, despite its reputation as one of the strongest algorithms for tabular data, performed marginally better at default settings than after tuning on this relatively small dataset, underscoring the importance of considering dataset size when designing hyperparameter search strategies.

The feature importance analysis provided a direct answer to the research question. Glucose is the strongest physiological predictor of diabetes in this dataset by a clear margin, followed by BMI, Age, and DiabetesPedigreeFunction. These findings align fully with established medical evidence and reinforce the value of targeting elevated glucose levels and obesity as the primary indicators in diabetes screening and prevention programmes.

Several directions for future work emerge from this study. Replacing row deletion with median or KNN imputation for handling missing values would preserve more data and potentially improve model performance, particularly for features such as Insulin and SkinThickness. Applying SMOTE or similar oversampling methods could better address class imbalance and improve recall. Exploring deep learning models or incorporating SHAP-based explainability tools would further enhance both predictive power and interpretability. Most importantly, validating the models on a diverse, multi-demographic clinical dataset would be an essential step towards any real-world application.

On the whole, this project demonstrates that machine learning, and ensemble boosting in particular, represents a practical and effective approach to early diabetes prediction. The ability to identify high-risk individuals based on routine clinical measurements has genuine potential to support preventive healthcare strategies and contribute to reducing the significant global burden that diabetes continues to impose.

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10. APPENDICES

Appendix A: Dataset Summary Statistics

Table A1: Descriptive Statistics of the Pima Indian Diabetes Dataset

Feature	Count	Mean	Std Dev	Min	25%	50%	75%	Max
Pregnancies	768	3.85	3.37	0.00	1.00	3.00	6.00	17.00
Glucose	768	120.89	31.97	0.00	99.00	117.00	140.25	199.00
BloodPressure	768	69.11	19.36	0.00	62.00	72.00	80.00	122.00
SkinThickness	768	20.54	15.95	0.00	0.00	23.00	32.00	99.00
Insulin	768	79.80	115.24	0.00	0.00	30.50	127.25	846.00
BMI	768	31.99	7.88	0.00	27.30	32.00	36.60	67.10
DiabetesPedigreeFunction	768	0.47	0.33	0.08	0.24	0.37	0.63	2.42
Age	768	33.24	11.76	21.00	24.00	29.00	41.00	81.00
Outcome	768	0.35	0.48	0.00	0.00	0.00	1.00	1.00

Appendix B: Zero Value Counts Per Feature

Table B1: Number of Medically Invalid Zero Values per Feature

Feature	Zero Value Count
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11

Appendix C: Hyperparameter Grids Used in GridSearchCV

Table C1: Hyperparameter Search Grids for Each Model

Gradient Boosting: `n_estimators` — [100, 200, 300]; `learning_rate` — [0.01, 0.05, 0.1]; `max_depth` — [2, 3, 4]; `subsample` — [0.8, 1.0].

XGBoost: `n_estimators` — [100, 200, 300]; `learning_rate` — [0.01, 0.05, 0.1]; `max_depth` — [3, 4, 5]; `subsample` — [0.8, 1.0]; `colsample_bytree` — [0.8, 1.0].

AdaBoost: `n_estimators` — [50, 100, 200, 300]; `learning_rate` — [0.01, 0.1, 0.5, 1.0].

Decision Tree: max_depth — [3, 5, 7, 10, None]; min_samples_split — [2, 5, 10]; min_samples_leaf — [1, 2, 4]; criterion — [gini, entropy].

Appendix D: Python Libraries Used

Table D1: Python Libraries and Their Purpose

NumPy — numerical computation and array operations. Pandas — data loading, manipulation, and analysis. Matplotlib — data visualisation and plot generation. Seaborn — advanced statistical visualisation. Scikit-learn — model building, preprocessing, evaluation, and hyperparameter tuning. XGBoost — XGBoost classification algorithm. Warnings — suppression of non-critical runtime warnings during model training.

Appendix E: Full Project Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV, StratifiedKFold,
cross_val_score, learning_curve

from sklearn.preprocessing import StandardScaler

from sklearn.ensemble import GradientBoostingClassifier, AdaBoostClassifier

from sklearn.tree import DecisionTreeClassifier

from xgboost import XGBClassifier

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
confusion_matrix, roc_curve, roc_auc_score

import warnings
warnings.filterwarnings("ignore")

# Load Data
df = pd.read_csv("diabetes.csv")

# Basic Data Inspection
print("Shape:", df.shape)
print(df.info())
print(df.describe())
print("Duplicates:", df.duplicated().any())
print("Null Values:", df.isnull().any())

# EDA - Plot 1: Outcome Distribution
outcome_counts = df['Outcome'].value_counts()
```



```

total_patients = outcome_counts.sum()
percentages = outcome_counts / total_patients * 100
labels = [f'0 - Non-Diabetic\n({outcome_counts[0]} / {percentages[0]:.1f} %)',
          f'1 - Diabetic\n({outcome_counts[1]} / {percentages[1]:.1f} %)']
colors = sns.color_palette("Set2", 2)
plt.figure(figsize=(8, 6))
plt.pie(outcome_counts, labels=labels, autopct='%1.1f%%', colors=colors,
        explode=(0.05, 0.05), shadow=True)
plt.title('Distribution of Diabetes Outcome')
plt.show()

# EDA - Plot 2: Feature Distributions
df.hist(figsize=(14,10), bins=20)
plt.suptitle('Feature Distributions')
plt.show()

# EDA - Plot 3: Boxplots by Outcome
features = ['Glucose','BMI','Age','Insulin']
plt.figure(figsize=(12,8))
for i, col in enumerate(features, 1):
    plt.subplot(2,2,i)
    sns.boxplot(x='Outcome', y=col, data=df)
    plt.title(f'{col} vs Outcome')
plt.tight_layout()
plt.show()

# EDA - Plot 4: Correlation Heatmap
plt.figure(figsize=(18,18))
sns.heatmap(df.corr(), linewidths=.5, annot=True)
plt.title('Correlation Matrix Heatmap')
plt.show()

# EDA - Plot 5: Pairplot
sns.pairplot(df[['Glucose','BMI','Age','Outcome']], hue='Outcome')
plt.show()

```

```

# EDA - Plot 6: Age vs Glucose Scatter Plot
plt.figure(figsize=(8,6))
sns.scatterplot(x='Age', y='Glucose', hue='Outcome', data=df)
plt.title('Age vs Glucose by Diabetes Outcome')
plt.show()

# EDA - Plot 7: Zero Value Analysis
zero_cols = ['Glucose','BloodPressure','SkinThickness','Insulin','BMI']
(df[zero_cols] == 0).sum().plot(kind='bar')
plt.title('Zero Values in Medical Features')
plt.ylabel('Count')
plt.show()

# Data Preprocessing
df[zero_cols] = df[zero_cols].replace(0, np.nan)
df = df.dropna()
print("Dataset shape after cleaning:", df.shape)

# Feature and Target Split
X = df.drop('Outcome', axis=1)
y = df['Outcome']

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

# Feature Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Model Building
gb_model = GradientBoostingClassifier(random_state=42)
gb_model.fit(X_train, y_train)

```

```

y_pred_gb = gb_model.predict(X_test)

xgb_model = XGBClassifier(use_label_encoder=False,
                           eval_metric='logloss', random_state=42)
xgb_model.fit(X_train, y_train)
y_pred_xgb = xgb_model.predict(X_test)

ada_model = AdaBoostClassifier(random_state=42)
ada_model.fit(X_train, y_train)
y_pred_ada = ada_model.predict(X_test)

dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
y_pred_dt = dt_model.predict(X_test)

# Model Evaluation Function
def evaluate_model(name, y_test, y_pred):
    print("Model:", name)
    print("Accuracy :", accuracy_score(y_test, y_pred))
    print("Precision:", precision_score(y_test, y_pred))
    print("Recall  :", recall_score(y_test, y_pred))
    print("F1 Score :", f1_score(y_test, y_pred))
    print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
    print("-"*50)

evaluate_model("Gradient Boosting", y_test, y_pred_gb)
evaluate_model("XGBoost", y_test, y_pred_xgb)
evaluate_model("AdaBoost", y_test, y_pred_ada)
evaluate_model("Decision Tree", y_test, y_pred_dt)

# Confusion Matrix Plots
def plot_confusion_matrix(y_true, y_pred, title):
    cm = confusion_matrix(y_true, y_pred)
    plt.figure(figsize=(5,4))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',

```

```

        xticklabels=['Non-Diabetic','Diabetic'],
        yticklabels=['Non-Diabetic','Diabetic'])
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title(title)
plt.show()

plot_confusion_matrix(y_test, y_pred_gb, "Gradient Boosting Confusion Matrix")
plot_confusion_matrix(y_test, y_pred_xgb, "XGBoost Confusion Matrix")
plot_confusion_matrix(y_test, y_pred_ada, "AdaBoost Confusion Matrix")
plot_confusion_matrix(y_test, y_pred_dt, "Decision Tree Confusion Matrix")

# Feature Importance
importance = ada_model.feature_importances_
imp_df = pd.DataFrame({'Feature': X.columns, 'Importance': importance})
imp_df = imp_df.sort_values(by='Importance', ascending=False)
plt.figure(figsize=(8,5))
sns.barplot(x='Importance', y='Feature', data=imp_df, palette="viridis")
plt.title("Feature Importance - AdaBoost")
plt.show()

# ROC Curve
y_prob = ada_model.predict_proba(X_test)[:,-1]
fpr, tpr, _ = roc_curve(y_test, y_prob)
auc_score = roc_auc_score(y_test, y_prob)
plt.figure(figsize=(6,5))
plt.plot(fpr, tpr, label="AdaBoost AUC = {:.3f}".format(auc_score))
plt.plot([0,1],[0,1], '--', color='gray')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - AdaBoost")
plt.legend(loc="lower right")
plt.grid(True)
plt.show()
print("AdaBoost AUC Score:", auc_score)

```

Hyperparameter Tuning

```
models = {  
    "Gradient Boosting": {  
        "estimator": GradientBoostingClassifier(random_state=42),  
        "params": {  
            'n_estimators':[100,200,300],  
            'learning_rate':[0.01,0.05,0.1],  
            'max_depth':[2,3,4],  
            'subsample':[0.8,1]  
        }  
    },  
    "XGBoost": {  
        "estimator": XGBClassifier(use_label_encoder=False,  
                                   eval_metric='logloss', random_state=42),  
        "params": {  
            'n_estimators':[100,200,300],  
            'learning_rate':[0.01,0.05,0.1],  
            'max_depth':[3,4,5],  
            'subsample':[0.8,1],  
            'colsample_bytree':[0.8,1]  
        }  
    },  
    "AdaBoost": {  
        "estimator": AdaBoostClassifier(random_state=42),  
        "params": {  
            'n_estimators':[50,100,200,300],  
            'learning_rate':[0.01,0.1,0.5,1]  
        }  
    },  
    "Decision Tree": {  
        "estimator": DecisionTreeClassifier(random_state=42),  
        "params": {  
            'max_depth':[3,5,7,10,None],  
            'min_samples_split':[2,5,10],
```

```

        'min_samples_leaf':[1,2,4],
        'criterion':['gini','entropy']
    }
}
}

results_list = []
for name, model_info in models.items():
    grid = GridSearchCV(model_info["estimator"], model_info["params"],
                        cv=5, scoring='f1', n_jobs=-1)
    grid.fit(X_train, y_train)
    best_model = grid.best_estimator_
    y_pred = best_model.predict(X_test)
    print(f"Best Parameters ( {name} ):", grid.best_params_)
    evaluate_model(name, y_test, y_pred)
    results_list.append({
        "Model": name,
        "Accuracy": accuracy_score(y_test, y_pred),
        "Recall": recall_score(y_test, y_pred),
        "F1 Score": f1_score(y_test, y_pred)
    })

results = pd.DataFrame(results_list).sort_values(
    by="F1 Score", ascending=False).reset_index(drop=True)
print("\nFinal Model Comparison")
print(results)

# Learning Curves
def plot_learning_curve(model, X, y, learning_rates, title):
    plt.figure(figsize=(15,5))
    for lr in learning_rates:
        model.set_params(learning_rate=lr)
        train_sizes, train_scores, val_scores = learning_curve(
            model, X, y, cv=5, scoring='f1', n_jobs=-1,
            train_sizes=np.linspace(0.1, 1.0, 5),

```

```

        shuffle=True, random_state=42)
train_mean = np.mean(train_scores, axis=1)
val_mean = np.mean(val_scores, axis=1)
plt.plot(train_sizes, train_mean, '--', label=f'Train lr={lr}')
plt.plot(train_sizes, val_mean, '-', label=f'Val lr={lr}')
plt.title(title)
plt.xlabel("Training Samples")
plt.ylabel("F1 Score")
plt.legend()
plt.grid(True)
plt.show()

gb_model = GradientBoostingClassifier(
    n_estimators=200, max_depth=3, subsample=1, random_state=42)
plot_learning_curve(gb_model, X_train, y_train,
                    learning_rates=[0.01, 0.05, 0.1],
                    title="Gradient Boosting Learning Curves")

xgb_model = XGBClassifier(
    n_estimators=200, max_depth=4, subsample=1, colsample_bytree=1,
    use_label_encoder=False, eval_metric='logloss', random_state=42)
plot_learning_curve(xgb_model, X_train, y_train,
                    learning_rates=[0.01, 0.05, 0.1],
                    title="XGBoost Learning Curves")

ada_model = AdaBoostClassifier(n_estimators=200, random_state=42)
plot_learning_curve(ada_model, X_train, y_train,
                    learning_rates=[0.01, 0.1, 0.5, 1],
                    title="AdaBoost Learning Curves")

```